Informational Actualization Model

A Technical Companion

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https://github.com/hmahaffeyges/IAM-Validation — Zenodo: 10.5281/zenodo.18702043

This companion document explains how every quantity in the IAM framework is derived and where the supporting measurements come from. It is intended as a readable guide for anyone encountering the model for the first time. Full mathematical derivations are in the IAM Theory Paper (see the final section).

The IAM Term

IAM adds a single term to the perturbation equations:

$$\beta_m \, E(a)$$

This term enters as additional friction in the matter growth equation, gently suppressing how quickly structure forms over cosmic time. Everything else in standard cosmology remains unchanged.

The two quantities in the term are:

$\beta_m = \Omega_m/2 = 0.1575$ — the informational coupling constant, derived from the virial theorem (Section 2).

$E(a) = \exp(1 - 1/a)$ — the activation function tracking cumulative informational entropy on the cosmic horizon, derived from the decoherence integral (Section 3).

The 9-step feedback loop that produces this term from first principles is:

gravity → decoherence → irreversible information → Landauer cost → holographic encoding → informational pressure → modified growth → feedback suppression → gravity

The loop closes back on itself — suppressed growth means less structure, less decoherence, less informational pressure — which is why the mechanism self-regulates rather than running away. Each arrow is an established physical result. The loop as a whole is IAM's contribution.

The Coupling Constant $\beta_m = \Omega_m/2$

For any self-gravitating system bound by a $1/r$ potential, the virial theorem requires:

$$2\langle T\rangle + \langle V\rangle = 0 \implies \langle T\rangle = \tfrac{1}{2}|\langle V\rangle|$$

Exactly half the gravitational energy budget resides in kinetic degrees of freedom. In IAM the kinetic channel is the decoherence-producing channel — irreversible random motions generate classical information requiring holographic encoding. The informational contribution to the energy density is therefore $\Omega_m/2$, giving $\beta_m = 0.1575$.

Real cosmological halos deviate from perfect virial equilibrium due to ongoing mergers, infalling material, and bulk angular momentum. The virial efficiency η_{virial} captures this, so the full expression is:

$$\beta_m = \Omega_m \times f_{\text{coll}} \times \eta_{\text{virial}}$$

where $f_{\text{coll}} = 0.62 \pm 0.03$ is the fraction of matter in collapsed structures [23]. Six independent N-body simulation groups have measured the mass-weighted virial ratio $2K/|U|$ directly: Bryan

& Norman (1998), Bett et al. (2007), Neto et al. (2007), Ludlow et al. (2010), Power et al. (2012), and Klypin et al. (2016). Their combined result is $\langle \eta_{\text{virial}} \rangle = 0.815 \pm 0.025$, giving:

$$\beta_m = 0.315 \times 0.62 \times 0.815 = 0.159 \pm 0.010$$

This agrees with the virial-theorem-derived $\Omega_m/2 = 0.1575$ to 1.1%. These measurements were made independently of IAM.

The 19% deviation of η_{virial} from unity has three well-characterized physical contributions: ongoing mergers and infall ($\sim$12%), ordered angular momentum from tidal torques ($\sim$4%), and non-thermal pressure support ($\sim$2%). Neto et al. [12] directly measure $\eta \approx 0.87$ for relaxed halos and $\eta \approx 0.72$ for actively merging halos; the mass-weighted combination gives $\eta \approx 0.81$. Power et al. [15] find the efficiency decreases by $\sim$5% from $z = 0$ to $z = 1$, consistent with halos being somewhat less settled at earlier times.

The same 1/2 partition holds across 37 orders of magnitude in physical scale — from hydrogen atoms (verified to machine precision using published ionization energies for all 20 elements H through Xe) through the Chandrasekhar stellar limit through galaxy clusters through the Planck MCMC posterior. This is not a coincidence; it is the virial theorem operating wherever a $1/r$ potential dominates in three spatial dimensions.

The Activation Function $E(a) = \exp(1 - 1/a)$

$E(a)$ is derived by integrating the decoherence rate over cosmic time, weighted by the Gibbons–Hawking encoding temperature $T_H = H/2\pi$ and the horizon area $A_H = 4\pi/H^2$. For the integral to produce a $-1/a$ dependence in the exponent, the structure formation rate must scale as $D(a)^n$ with an effective exponent $n_{\text{eff}} = 5/2$ in the analytical matter-dominated limit, shifting to $n_{\text{eff}} \approx 3.0$–$3.5$ in the full ΛCDM history. Six published structure-formation analyses find $n_{\text{eff}} = 3.22 \pm 0.44$, consistent with this range. The Sheth–Tormen mass function at galaxy-scale halos ($\sigma_* = 1.2$) recovers the $1/a$ exponent to within 1% [20].

Two boundary conditions fix $E(a)$ completely without any free parameters. The normalization $E(1) = 1$ sets the present-day contribution equal to the matter energy density scale. The saturation limit $E(a) \to e$ as $a \to \infty$ follows from the self-limiting nature of the expansion feedback: as expansion dilutes matter, the decoherence rate slows and the cumulative entropy approaches a finite ceiling.

At $a = 1$ today, $E(1)/e = 1/e = 36.8\%$, placing the current epoch at the inflection point of the activation function where the rate of informational entropy production is at its peak.

The Sector Split: $\mu < 1$ and $\Sigma = 1$

IAM predicts that matter and photons experience different effective gravitational couplings, described by the standard μ–Σ parameters where μ modifies the Poisson equation for matter and Σ modifies the lensing potential. The origin of the split is geometric.

Gravitational decoherence requires time to act — a quantum state needs a finite proper time interval to become classical [26]. Massive particles have that time: they move through space on timelike paths, continuously interacting gravitationally as they collapse and cluster. Photons do not: they travel on null paths where no proper time elapses, so they are never subject to this process. The prediction $\Sigma = 1$ follows from this geometric fact, not from a parameter choice.

The specific value $\mu_0 = -0.136$ is calculated directly from β_m, $E(a)$, and the Planck 2018 cosmological parameters. It measures how much the matter growth rate is suppressed relative to standard cosmology by the additional friction term $\beta_m E(a)$ at the present epoch $a = 1$. No parameter was adjusted to produce this value.

The combination $\mu < 1$ with $\Sigma = 1$ is not predicted by other modified gravity frameworks. $f(R)$ gravity gives $\mu > 1$, $\Sigma > 1$. DGP braneworld gives $\mu > 1$. General Horndeski theories generally predict $\Sigma \neq 1$. The IAM signature in μ–Σ space is unique.

The H_0 Sector Split

A direct consequence of the dual-sector structure is that photon-sector probes and matter-sector probes measure different effective Hubble parameters. This is not a tension to be resolved — it is a prediction.

Photon-sector probes (CMB, BAO, standard candles calibrated via the photon ladder) measure the background expansion rate:

$$H_0^\gamma = 67.4\ \mathrm{km\,s^{-1}Mpc^{-1}}$$

Matter-sector probes (Cepheid distance ladder, TRGB, gravitational wave sirens) measure the effective expansion rate experienced by matter, which includes the $\beta_m E(a)$ term:

$$H_0^m = H_0^\gamma \sqrt{1+\beta_m} = 67.4 \times \sqrt{1.1575} = 72.5\ \mathrm{km\,s^{-1}Mpc^{-1}}$$

Both values are correct measurements of real physical quantities in different sectors. The Level 2 MCMC posterior gives $H_0^{\mathrm{matter}} = 72.26\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$, 0.75σ from SH0ES [19], with zero free parameters.

What "Zero Free Parameters" Means

IAM is described as a physically motivated extension with no new free amplitudes. This is worth unpacking precisely:

- $\beta_m = \Omega_m/2$: derived from the virial theorem. Its value changes only if Ω_m changes. It was not fitted to any cosmological observable.
- $E(a) = \exp(1 - 1/a)$: derived from the decoherence integral. Both boundary conditions ($E(1) = 1$ and $E(\infty) = e$) follow from the physics. No amplitude, shape parameter, or transition redshift was adjusted.
- $\mu_0 = -0.136$: calculated directly from β_m and Planck 2018 parameters. It is an output of the model, not an input.
- $H_0^{\mathrm{matter}} = 72.5$: calculated from H_0^γ and β_m. Again an output.

The model has the same number of free parameters as ΛCDM. The IAM term $\beta_m E(a)$ adds no new degrees of freedom because both β_m and $E(a)$ are fully determined by established physics and the standard cosmological parameters already constrained by Planck.

The Thermodynamic Foundation

The IAM causal chain rests on a sequence of established results, each with independent experimental or theoretical support:

- **Gravitational decoherence as the source of irreversibility:** Zurek [26] and Penrose [13].
- **Information has thermodynamic cost (Landauer's principle):** $k_B T \ln 2$ per bit. Experimentally verified by Bérut et al. [2] and Jun et al. [9].
- **Holographic encoding on the horizon:** Bekenstein [1], 't Hooft [24], Susskind [22].
- **Entropy functionals determine gravitational field equations:** Jacobson [7]; extended to FRW cosmology by Cai & Kim [5].

IAM's single new identification is that decoherence-produced information contributes a specific term S_{info} to the horizon entropy functional, entering the perturbation equations through the Cai–Kim first law as $S_{\mathrm{total}} = S_{\mathrm{geo}} + S_{\mathrm{info}}$.

The MCMC Validation

IAM has been validated against the Planck 2018 CMB likelihood through 17 converged MCMC chains. 12 Level 1 chains used MGCAMB's μ–Σ parametrization across multiple dataset combinations. 3 Level 2 chains used direct modification of CAMB's Fortran perturbation equations implementing the physical dual-sector mechanism. 2 additional exploratory chains tested whether the IAM term should instead be applied upstream in the background rather than the perturbation equations; those chains converged cleanly but returned $H_0 \approx 61.5\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, confirming the mechanism belongs in the perturbation equations, not the background. All 17 chains satisfy $R - 1 < 0.01$ Gelman–Rubin convergence. All chains, YAML configurations, and analysis scripts are publicly available in the repository.

The fit to the full Planck 2018 likelihood gives $\Delta\chi^2 = +0.54$ relative to ΛCDM. This is the expected behavior: IAM modifies perturbations at late times ($z < 2$) and does not alter the CMB power spectrum, BAO positions, or photon transfer functions, which are set at recombination.

The σ_8 suppression from 0.809 to 0.800 was verified to arise from real growth physics. After the Level 2 chains converged, $\ln A_s$ shifted by 0.06σ and Ω_m shifted by 0.04σ — well within statistical fluctuation, not systematic degeneracy.

When comparing chains evaluated against different likelihood sets, the apples-to-apples procedure evaluates the ΛCDM posterior on any additional likelihoods separately before computing $\Delta\chi^2$. The script `rsd_apples_to_apples.py` in the repository implements this.

Falsifiable Predictions

IAM makes three near-term predictions that are quantitative and directly testable with existing or imminent data.

Euclid μ–Σ measurement. The IAM prediction $\mu_0 = -0.136$, $\Sigma_0 = 0$ is directly testable by Euclid at projected $\sigma(\mu_0) \sim 0.04$ sensitivity. If Euclid measures μ consistent with zero at this precision, IAM is ruled out. The combination $\mu < 1$ with $\Sigma = 1$ is a unique signature in the μ–Σ plane that distinguishes IAM from all other current modified gravity proposals: $f(R)$ gravity gives $\mu > 1$, $\Sigma > 1$; DGP braneworld gives $\mu > 1$; general Horndeski theories predict $\Sigma \neq 1$.

DESI DR2 growth rate trajectory. DESI DR2 (2025) shows a 2.8–4.2σ preference for dynamical dark energy in $w_0 w_a$ fits, with an apparent phantom crossing at $z \sim 0.5$. IAM predicts this signal is not a real phantom crossing — it is a growth-geometry tension that $w_0 w_a$ fits are absorbing. The IAM $f\sigma_8(z)$ trajectory, suppressed by the $\beta_m E(a)$ friction term, is consistent with the DESI DR2 growth rate data without requiring $w < -1$. A direct confrontation of the DESI DR2 $f\sigma_8$ measurements against the IAM predicted trajectory tests this interpretation.

Galaxy cluster lensing-to-dynamics ratio. The dual-sector structure predicts $M_{\mathrm{lens}}/M_{\mathrm{dyn}} = 1/\mu_{\mathrm{eff}}(z) > 1$ in a redshift-dependent pattern that approaches unity at high z as $\mu(a) \to 1$. This is qualitatively distinct from a constant hydrostatic bias and testable with current eROSITA, Planck SZ, and weak lensing datasets in combination.

Future Data

Several near-term datasets will help further refine IAM's predictions:

- **Euclid (∼2026–2030):** Independent measurement of μ and Σ from the same survey volume, refining the amplitude of the sector split and the redshift dependence of $\mu(z)$.
- **DESI Year 5:** Growth rate $f\sigma_8(z)$ at ∼1% precision across $0 < z < 2$, constraining the shape of the suppression ramp predicted by $E(a)$.
- **E_G statistic:** Next-generation lensing surveys will sharpen the measurement of $E_G = \Omega_m \Sigma/\mu$, currently $\approx 0.40 \pm 0.03$ from BOSS [18], above the GR prediction of 0.315.

- **eROSITA × Planck SZ × weak lensing:** Three-way cluster ratios across large samples will constrain the sector structure at group and cluster scales.
- **CMB-S4:** Precision constraints on any photon-sector modification, further characterizing the sector boundary.

Key Numbers at a Glance

Quantity	IAM Value	Current Data	Source
μ_0	-0.136 (derived)	0.05 ± 0.22	DESI 2025
Σ_0	0 (null worldlines)	0.008 ± 0.045	ACT+WMAP+SDSS
σ_8	0.800 (derived)	0.802 ± 0.020	Stölzner et al. 2025
H_0^{matter}	72.5 km/s/Mpc	73.04 ± 1.04	Riess et al. 2022
$\Delta\chi^2$	+0.54 (late-time mechanism)	+0.54 confirmed	17 Planck chains
β_m	$\Omega_m/2 = 0.1575$	0.159 ± 0.010	6 N-body studies
$\langle\eta_{\text{virial}}\rangle$	0.81 (derived)	0.815 ± 0.025	6 N-body studies
n_{eff}	3.0–3.5 (derived)	3.22 ± 0.44	6 SF analyses
E_G	> 0.315 ($\mu < 1$)	$\approx 0.40 \pm 0.03$	BOSS + lensing
β_m/Ω_m	1/2 (virial theorem)	1/2 to machine precision	37 orders

For Full Derivations

This companion covers the key quantities and their origins at a summary level. The complete mathematical derivations — including the full Jacobson thermodynamics, the Cai–Kim extension to FRW cosmology, the introduction of S_{info}, the perturbation-level sector split, the variational formulation, and all supporting calculations — are in:

IAM Theory Paper: Horizon Thermodynamics and Gravitational Decoherence as the Origin of $\mu < 1$, $\Sigma = 1$
Heath W. Mahaffey, 2026
Available in the `/docs` folder of the repository and via the Zenodo DOI above.

Paper Portfolio

The following papers are available in the `/docs` folder of the repository and via the Zenodo DOI.

IAM Theory Paper — Complete derivation chain: Jacobson (1995) → Cai–Kim (2005) → IAM. Derives $E(a)$, β_m, the μ–Σ sector split, and the perturbation-level sector separation from first principles.

IAM Master Preprint — The consolidated observational and theoretical document. Full MCMC validation record, the dual-sector H_0 split, and the σ_8 suppression result.

Dual-Sector Validation Paper — Empirical validation of the sector separation using Pantheon+ Type Ia supernovae through three independent tests (Planck prior, SH0ES prior, no prior).

Dual-Sector Perturbation Cosmology: CAMB — Documents the Level 2 direct Fortran implementation of the dual-sector mechanism in CAMB's perturbation equations. The physical paper behind the modified `equations.f90` and the three converged Level 2 chains ($\Delta\chi^2 = +0.54$ best-fit under full Planck 2018 likelihood).

Late-Time Growth Suppression in the μ–Σ Framework — Confrontation with Planck and large-scale structure.

Dark Energy or Sector Tension? — Positions IAM within the DESI DR2 dynamical dark energy discussion. Argues the apparent phantom crossing is a growth-geometry tension, not a dark energy signature.

IAM Lensing and Dynamics Paper — Derives the cluster discriminator $M_{\rm lens}/M_{\rm dyn} = 1/\mu_{\rm eff}(z)$ and its redshift dependence. Falsification signatures for eROSITA and Euclid cluster samples.

3-Way Mass Discrepancy in Galaxy Clusters — SZ/X-ray/lensing mass ordering tests showing $M_{\rm lens} > M_{\rm SZ} \approx M_{\rm hydro}$, separating IAM from pure hydrostatic bias explanations via three-way internal consistency.

IAM Survey Predictions Paper — Pre-registered falsification roadmap for Euclid, DESI Year 5, and CMB-S4, with conservative exclusion criteria for the IAM baseline prediction $\mu_0 = -0.136$.

Virial Efficiency and Effective Nonlinear Exponent — N-body literature consistency closure for the virial coupling derivation. Shows the required $\eta_{\rm virial}$ and $n_{\rm eff}$ are consistent with six independent published simulation analyses, directly supporting the "derived parameter" status of $\beta_m = \Omega_m/2$.

Virial Partition Across Wide Domains — Demonstrates that the equal 1/2 energy partition holds across 37 orders of magnitude in physical scale, from atomic systems through stellar endpoints through galaxy clusters, wherever a $1/r$ potential dominates in three spatial dimensions.

IAM M-σ Paper — Derives $M_{\rm BH} = 2.00 \times 10^8 (\sigma/200)^4\, M_\odot$ from first principles using the same virial-partition logic that underlies β_m, matching observations to 2%. Connects the cosmological IAM mechanism to astrophysical structure scaling.

IAM Black Hole Thermodynamics — Extends the horizon-encoding logic to black hole horizons, clarifying what "encoding" means at BH boundaries and its relationship to the cosmological IAM mechanism.

IAM Black Hole Information Paradox — Applies IAM information bookkeeping to the black hole information paradox using the same holographic boundary framework as the cosmological derivation.

IAM CAMB Technical Note — Full technical documentation of the Level 1 and Level 2 MCMC implementations, modified Fortran source, and the apples-to-apples comparison procedure.

Extension papers covering gravitational decoherence at the quantum level, the quantum measurement problem, the electron mass, and far-future cosmology are maintained separately and are available in the same repository.

References

[1] Bekenstein, J. D., 1973, "Black holes and entropy," *Phys. Rev. D*, 7, 2333.

[2] Bérut, A., et al., 2012, "Experimental verification of Landauer's principle," *Nature*, 483, 187.

[3] Bett, P., et al., 2007, "The spin and shape of dark matter haloes in the Millennium simulation," *MNRAS*, 376, 215.

[4] Bryan, G. L. & Norman, M. L., 1998, "Statistical properties of X-ray clusters," *ApJ*, 495, 80.

[5] Cai, R.-G. & Kim, S. P., 2005, "First law of thermodynamics and Friedmann equations of FRW universe," *JHEP*, 02, 050.

[6] Clausius, R., 1870, "On a mechanical theorem applicable to heat," *Phil. Mag.*, 40, 122.

[7] Jacobson, T., 1995, "Thermodynamics of spacetime: The Einstein equation of state," *Phys. Rev. Lett.*, 75, 1260.

[8] Jenkins, A., et al., 2001, "The mass function of dark matter haloes," *MNRAS*, 321, 372.

[9] Jun, Y., Gavrilov, M., & Bechhoefer, J., 2014, "High-precision test of Landauer's principle," *Phys.*

Rev. Lett., 113, 190601.

[10] Klypin, A., et al., 2016, "MultiDark simulations," *MNRAS*, 457, 4340.

[11] Ludlow, A. D., et al., 2010, "The formation of ΛCDM haloes from the inside out," *MNRAS*, 406, 137.

[12] Neto, A. F., et al., 2007, "The statistics of ΛCDM halo concentrations," *MNRAS*, 381, 1450.

[13] Penrose, R., 1989, *The Emperor's New Mind*, Oxford University Press.

[14] Planck Collaboration, 2020, "Planck 2018 results VI," *A&A*, 641, A6.

[15] Power, C., et al., 2012, "Spurious evolution in cosmological N-body simulations," *MNRAS*, 419, 1576.

[16] Press, W. H. & Schechter, P., 1974, "Formation of galaxies and clusters," *ApJ*, 187, 425.

[17] Reed, D. S., et al., 2003, "Evolution of the mass function of dark matter haloes," *MNRAS*, 346, 565.

[18] Reyes, R., et al., 2010, "Confirmation of general relativity on large scales," *Nature*, 464, 256.

[19] Riess, A. G., et al., 2022, "A comprehensive measurement of the local value of the Hubble constant," *ApJL*, 934, L7.

[20] Sheth, R. K. & Tormen, G., 1999, "Large-scale bias and the peak background split," *MNRAS*, 308, 119.

[21] Stölzner, B., et al., 2025, "KiDS-Legacy: Consistency of cosmic shear measurements and joint cosmological constraints with external probes," *A&A*, arXiv:2503.19442.

[22] Susskind, L., 1995, "The world as a hologram," *J. Math. Phys.*, 36, 6377.

[23] Tinker, J., et al., 2008, "Toward a halo mass function for precision cosmology," *ApJ*, 688, 709.

[24] 't Hooft, G., 1993, "Dimensional reduction in quantum gravity," arXiv:gr-qc/9310026.

[25] Watson, W. A., et al., 2013, "The halo mass function through the cosmic ages," *MNRAS*, 433, 1230.

[26] Zurek, W. H., 2003, "Decoherence, einselection, and the quantum origins of the classical," *Rev. Mod. Phys.*, 75, 715.